第二次作业（10月14日课间交、共6题）： 课本正文第78页的2、3、4、6题，第84页的题2，第95页的题12.

78-2

Suppose $\{A_n\}$ are disjoint measurable sets, $B_n \subset A_n$, prove that

$$m^*\left(\bigcup B_n\right) = \sum_{n=1}^{\infty} m^*(B_n)$$

Proof: Recall $m^*\left(\bigcup B_n\right) \leqslant \sum m^*(B_n)$. Let $B = \bigcup B_n$, then A_n measurable implies $m^*(B) = m^*(B_n) + m^*(B \setminus A_n)$. By induction we can show that $m^*(B) = \sum_{k=1}^{n} m^*(B_k) + m^*\left(\bigcup_{k>n} B_k\right)$, hence $m^*(B) \geqslant \sum m^*(B_n)$. Therefore

$$m^*\left(\bigcup B_n\right) = \sum m^*(B_n).$$

78-3

Suppose $E_1, E_2 \subset \mathbb{R}^n$ where E_1 is measurable and $E_1 \Delta E_2$ is null, prove that E_2 is measurable and $m(E_2) = m(E_1)$.
Proof: Let $N = E_1 \Delta E_2$, then $E_2 = E_1 \cup (N \cap E_2) \setminus (N \cap E_1)$, where $N \cap E_2$, $N \cap E_1$ are null sets, so E_2 is measurable and $m(E_2) = m(E_1)$.

78-4

Suppose $B \subset \mathbb{R}^n$ satisfy for any $\varepsilon > 0$ there is a measurable A, such that $m^*(A \Delta B) < \varepsilon$, prove that B is measurable.
Proof: Take A_n measurable such that $m^*(A_n \Delta B) < 2^{-n}$. Let $B_n = \bigcup_{k>n} A_k$, then $B_n \Delta B \subset \bigcup (A_k \Delta B)$, so $m^*(B_n \Delta B) < 2^{1-n}$. Analogously, $\left(\bigcap B_n\right) \Delta B \subset \bigcup_{k>N} (A_k \Delta B)$, so $m^*\left(\left(\bigcap B_n\right) \Delta B\right) = 0$. Clearly $\bigcap B_n$ is measurable, hence B is also measurable.

78-6

Suppose $X = \{E_\alpha\}$ is a collection of disjoint subsets of $\mathbb{R}$ with positive measure, prove that X is countable.
Proof: For each n, let $I_{n,j} = \left\{\alpha : m^*(E_\alpha \cap [-n, n]) > \frac{1}{j}\right\}$. Clearly each I_n is finite, since $m([-n, n])$ is finite, and $\bigcup I_{n,j}$ contains all the indices, so X is countable.

84-2

Suppose $A \in \mu$, $B \subset \mathbb{R}^n$, prove that

$$m^*(A \cup B) + m^*(A \cap B) = m^*(A) + m^*(B).$$

Proof: By definition,

$$m^*(A \cup B) = m^*(A) + m^*((A \cup B) \setminus A)$$
$$m^*(B) = m^*(A \cap B) + m^*(B \setminus A)$$

which implies $m^*(A \cup B) + m^*(A \cap B) = m^*(A) + m^*(B)$.

95-12

Suppose $\{B_k\}$ is a sequence of decreasing measurable sets in $\mathbb{R}^n$, $m^*(A) < \infty$. Let $E_k = A \cap B_k$ and $E = \bigcap E_k$, prove that

$$\lim_{k\to\infty} m^*(E_k) = m^*(E).$$

Proof: Since $E \subset E_k$, $m^*(E_k) \geqslant m^*(E)$ so $\lim_{k\to\infty} m^*(E_k) \geqslant m^*(E)$. Let $C_k = E_k \setminus E_{k+1}$, then as subsets of measurable sets $B_k \setminus B_{k+1}$, $\sum_{n=1}^{\infty} m^*(C_n) = m^*\left(\bigcup C_n\right) < \infty$. Therefore $m^*(E_n) - m^*(E) = \sum_{k\geqslant n} m^*(C_k) \to 0$.